केंद्रीय विद्यालय संगठन, कोलकाता संभाग

KENDRIYA VIDYALAYA SANGATHAN, KOLKATA REGION

प्री-बोर्ड परीक्षा/ PRE-BOARD EXAMINATION- 2025-26

कक्षा / CLASS - X   अधिकतम अंक / MAX. MARKS - 80

विषय / SUB. - MATHEMATICS   समय / TIME - 3 घंटे/ hours

General Instructions

Read the following instructions carefully and follow them:

1. This question paper contains 38 questions. All Questions are compulsory.

2. This Question Paper is divided into 5 Sections A, B, C, D and E.

3. In Section A, Question numbers 1-18 are multiple choice questions (MCQs) and question no.19 and 20 are Assertion- Reason based questions of 1 mark each.

4. In Section B, Question numbers 21-25 are very short answer (VSA) type questions, carrying 02 marks each.

5. In Section C, Question numbers 26-31 are short answer (SA) type questions, carrying 03 marks each.

6. In Section D, Question numbers 32-35 are long answer (LA) type questions, carrying 05 marks each.

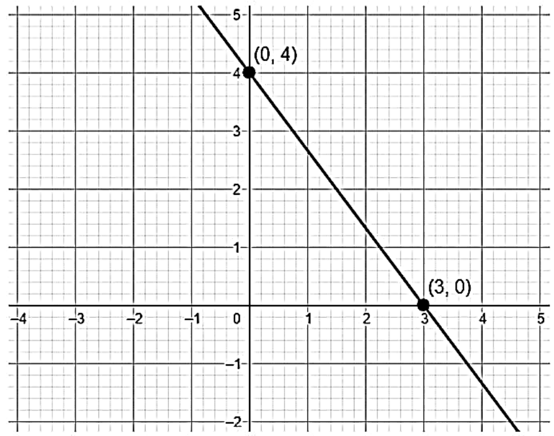
7. In Section E, Question numbers 36-38 are case study-based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.

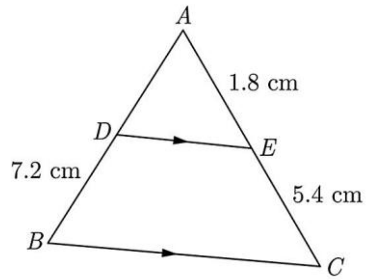
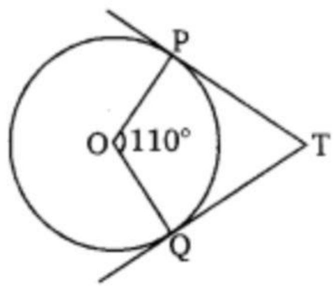
8. There is no overall choice. However, an internal choice in 2 questions of Section B, 2 questions of Section C and 2 questions of Section D has been provided. An internal choice has been provided in all the 2 marks questions of Section E.

9. Draw neat and clean figures wherever required. Take = 22/7wherever required if not stated.

10. Use of calculators is not allowed.

	Section A	
1.	If $a = 2^2 \times 3$, $b = 2 \times 5$, $c = 3^2 \times 7$, then LCM (a,b,c) is (a)1260 (b) 210 (c) 180 (d) None of these	1
2.	The given linear polynomial $y = f(x)$ has	1

	 (a) 2 zeros (b) 1 zero and the zero is ‘3’ (c) 1 zero and the zero is ‘4’ (d) No zero	
3.	The pair of equations y = 0 and y = -7 has: (a) One solution (b) Two solutions (c) Infinitely many solutions (d) No solution	1
4.	If $x^2 + 5kx + 16 = 0$, has equal roots, then the value of k is (a) ±64/25 (b) ±8/5 (c) ±5/8 (d) ±25/64	1
5.	Two APs have the same common difference. The first term of one of these is –1 and that of the other is – 8. Then the difference between their 4th terms is (a) 1 (b) 2 (c) 7 (d) 9	1
6.	The value of x for which 2x, x+10 & 3x+2 are the three consecutive terms of an AP is (a) 6 (b) -6 (c) -2 (d) 2	1
7.	In △ABC, it is given that AB = 9 cm, BC = 6 cm and CA = 7.5 cm. Also,△DEF is given such that EF = 8 cm and△DEF ~△ABC. Then, perimeter of△DEF is (a) 30 cm (b) 22.5 cm (c) 27 cm (d) 25 cm	1
8.	If in ΔABC and ΔDEF, AB/EF=AC/DE then they will be similar when (a) ∠A = ∠D (b) ∠A = ∠E (c) ∠C = ∠F (d) ∠B = ∠E	1
9.	In Figure, DE || BC . Find the length of side AD, given that AE = 1.8 cm BD = 7.2 cm and CE = 5.4 cm.	1

	 (a) 2.4 cm (b) 2.2 cm (c) 3.2 cm (d) 3.4 cm	
10.	In figure, if TP and TQ are the two tangents to a circle with centre O so that $\angle POQ = 110°$, then $\angle PTQ$ is equal to  (a) 60° (b) 70° (c) 80° (d) 90°	1
11.	If O is the centre of a circle and Chord PQ makes an angle 50° with the tangent PR at the point of contact P, find the angle made by the chord at the centre. (a) 130° (b) 100° (c) 50° (d) 30°	1
12.	From an external point P, tangents PA and PB are drawn to a circle with centre O. If CD is the tangent to the circle at a point E and PA = 14 cm. The perimeter of ΔPCD is (a) 14 cm (b) 21 cm (c) 28 cm (d) 35 cm	1
13.	$(\sec A + \tan A)(1 - \sin A) =$ (a) sec A (b) sin A (c) cosec A (d) cos A	1
14.	$9\sec^2 A - 9\tan^2 A =$ (a) 1 (b) 9 (c) 0 (d) 8	1
15.	If a circular grass lawn of 35 m in radius has a path 7 m wide running around it on the outside, then the area of the path is (a) 1450 m^2 (b) 1576 m^2 (c) 1694 m^2 (d) 3368 m^2	1
16.	The cumulative frequency table is useful in determining (a) Mean (b) Median (c) Mode (d) All of these	1
17.	Let empirical relationship between the three measures of central tendency be	1

	a(Median) = Mode + b(Mean), then (2b + 3a) = (a) 11 (b) 12 (c) 13 (d) 14	
18.	Harpreet tosses two different coins simultaneously. What is the probability that she gets at least one head ? (a) ¼ (b) ½ (c) ¾ (d) 1	1
19.	**Assertion (A):** Card numbered as 1, 2, 3............15 are put in a box and mixed thoroughly, one card is then drawn at random. The probability of drawing an even number is 7/15. **Reason (R):** For any event E, we have $0 \leq P(E) \leq 1$. (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A) (b) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A) (c) Assertion (A) is true but reason (R) is false. (d) Assertion (A) is false but reason (R) is true.	1
20.	**Assertion (A) : PA and PB are two tangents to a circle with centre O. Such that $\angle AOB = 110°$, then $\angle APB = 90°$.** **Reason (R): The length of two tangents drawn from an external point are equal.** (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A). (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A). (c) Assertion (A) is true but reason (R) is false. (d) Assertion (A) is false but reason (R) is true.	1
	Section B	
21.	(a) Explain whether $4 \times 11 \times 13 + 4$ is a prime number or a composite number. OR (b) Check whether 4^n can end with the digit 0 for any natural number n.	2
22.	If α and β are zeroes of $x^2 - (k - 6)x + 2(2k - 1)$, find the value of k if $\alpha + \beta = \alpha\beta/2$.	2
23.	Find the nature of the roots of the equation $3x^2 - 4\sqrt{3}x + 4 = 0$	2
24.	(a) Evaluate: $\sin^2 60^\circ + 2\tan 45^\circ - \cos^2 30^\circ$ OR (b) If sin x = 7/25, where x is an acute angle then find the value of sinx cosx (tanx + cotx)	2

25.	Find out the point which divides the line segment joining the points (7, -6) and (3, 4) in ratio 1 : 2 internally.	2
	Section C	
26.	Prove that $3 + 2\sqrt{5}$ is an irrational number.	3
27.	Ramkali saved Rs 5 in the first week of a year and then increased her weekly saving by Rs 1.75. If in the nth week, her weekly savings become Rs 20.75, find n.	3
28.	Prove that : $\tan\theta/(1-\cot\theta) + \cot\theta/(1-\tan\theta) = 1 + \sec\theta \operatorname{cosec}\theta$	3
29.	(a) Solve: $\frac{2}{\sqrt{x}} + \frac{3}{\sqrt{y}} = 2$; $\frac{4}{\sqrt{x}} - \frac{9}{\sqrt{y}} = -1$ OR (b) A fraction becomes 9/11 , if 2 is added to both the numerator and the denominator. If, 3 is added to both the numerator and the denominator it becomes 5/6. Find the fraction.	3
30.	A quadrilateral ABCD is drawn to circumscribe a circle. Prove that AB + CD = AD + BC	3
31.	(a) The frequency distribution table of agriculture holdings in a village is given below:	3

Area of land (in hectares)	1-3	3-5	5-7	7-9	9-11	11-13
No. of families	20	45	80	55	40	12

Find the modal agriculture holdings of the village.

OR

(b) If the mean of the following distribution is 54, find the value of p.

Class Interval	0-20	20-40	40-60	60-80	80-100
Frequency	7	p	10	9	13

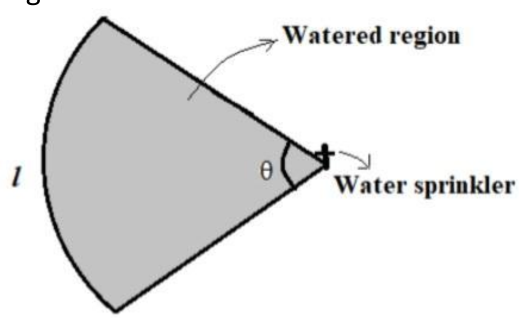
	Section D	

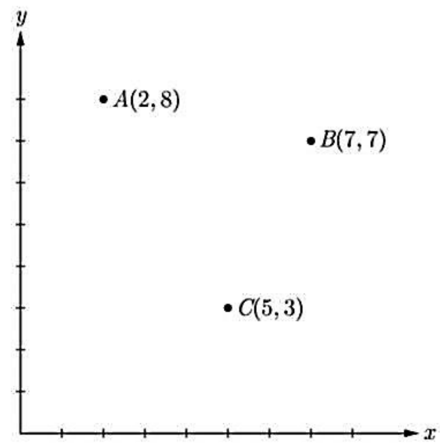
32.	A train travels 360 km at a uniform speed. If the speed had been 5km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train.	5
33.	State & prove BASIC PROPORTIONALITY THEOREM.	5
34.	(a) The person standing on the bank of the river observes that the angle of elevation of the top of a tree standing on the opposite bank is 60°. When he moves 30 m away from the bank, he finds the angle of elevation to be 30°. Find the height of the tree and the width of the river. OR (b) As observed from the top of a 100 m high light house from the sea-level, the angles of depression of two ships are 30° and 45°. If one ship is exactly behind the other on the same side of the light house, find the distance between the two ships.	5
35.	(a) A cubical block of side 7 cm is surmounted by a hemisphere. What is the greatest diameter the hemisphere can have? Find the surface area of the solid. OR (b) A gulab jamun contains sugar syrup up to about 30% of its volume. Find approximately how much syrup would be found in 45 gulab jamuns, each shaped like a cylinder with two hemispherical ends with a length of 5 cm and a diameter of 2.8 cm.	5

Section E

36. One of four main blood types can be found in a human body. They are known as A, B, AB and O. Each blood type can be further classified as either a Rhesus positive (+) or Rhesus negative (−).For example, a possible combination is blood type O and Rhesus negative which is written as O^-
The data below shows the distribution of the blood types and Rhesus types of given blood type for a Blood Donation Center recorded (in percentages) for the year 2023.

BLOOD GROUP	RHESUS FACTOR	NUMBER OF PERSONS (in %)
O	O^-	x
	O^+	30
A	A^-	8
	A^+	24
B	B^-	6
	B^+	18
AB	AB^-	1
	AB^+	3

	i) Find the value of . ii) Find the probability that a randomly selected person has a Rhesus negative blood type. iii) (A) What is the probability that the person selected from the record is Rhesus positive but neither blood type A nor B? OR (B) People with blood type AB positive (AB+) are known as the universal recipient and with blood type O negative (O−) are known as universal donor. Find the probability of a selected person to be neither universal recipient nor universal donor.	1 1 2
37.	A water sprinkler is a device used to irrigate agricultural crops, lawns, landscapes, golf courses, and other areas. Water sprinklers can be used for residential, industrial, and agricultural usage. A water sprinkler is set to shoot a stream of water a distance of 21 m and rotate through an angle which is equal to complementary angle of 10°.  i) What is the area of sector in terms of arc length? ii) What is the area of the watered region (in terms of)? iii) (A) If the radius(r) changes to 28m, find the angle θ so that the area of the watered region remains the same. OR (B) If the radius(r) is increased from 21m to 28m and the angle remains the same, what is the increase in the area of the watered region?	1 1 2
38.	Three friends Amar, Bandhu and Chakradev lives in societies represented by the points A, B and C respectively. They all work in offices located in a same building represented by the point O (origin). Since they all go to same building everyday, they decided to do carpooling to save money on petrol. Based on the above information, answer the following questions.	

	 (i) Which society is nearest to the office? (ii) What is the distance between A and C ? (iii) (A)Find the least distance between AB , OA and BC ? OR (B) If Bandhu and Chakradev planned to meet at a club situated at the mid-point of the line joining the points B and C , find the coordinates of this point.	1 1 2

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